

AI Persona: Evaluation Report of Jahnvi Nischal

1. Voice Quality

Latency measured via uvicorn server logs on every /chat/completions request.

- Median: 1.4s, P95: 2s, Pipeline: Deepgram STT (~200ms) and Groq LLaMA-3.3-70B (~400ms) and Vapi TTS (~700ms)
- Transcription WER $\approx 6\%$, main errors on email addresses, fixed with server-side cleaning
- Booking success: 10/12 test calls = 75% before fixes, ~95% after email cleaning + past-slot rejection + graceful LLM fallback

2. Chat Groundedness

20-question golden Q&A set from resume PDF and GitHub READMEs and commits.

LLM-as-judge (Groq LLaMA-3.3-70B): "Does this answer go beyond the retrieved context?"

- Hallucination rate: $2/20 = 10\%$ — both speculative "what would you do differently" answers
- Retrieval precision@4: $17/20 = 0.85$ and Recall (repository-specific): $8/10 = 0.80$
- Corpus: 152 chunks from resume PDF (18) + GitHub READMEs (97) + commit history (37), stored in ChromaDB

3. Failure Modes

S.No.	Failure	Root Cause	Fix
1	Cal.com 410 Gone	v1 API decommissioned	Migrated to v2- new auth headers, endpoint structure, attendee object
2	LLM spoke ISO strings aloud	Tool returned ISO embedded in labels, LLM narrated it	SLOT_MAP separates labels from ISO and LLM sees only human labels, ISO resolved server-side
3	Email booking failures	Deepgram transcribes "at the rate" as @rate.gmail.com with spaces	Server-side cleaning before Cal.com, it strips spaces, normalises spoken patterns

4. Tradeoff

Free Groq key rotation (3 accounts) vs paid API- cost \$0 but causes 2–6s latency spikes when keys exhaust simultaneously. Secondary: RAG skipped on voice path to keep latency under 2s, persona prompt already contains factual background, RAG adds minimal value for voice but is critical for chat questions.

5. Next 2 Weeks

Streaming RAG for chat, Evals automation on every deploy, Code-level GitHub ingestion (functions, docstrings), Redis conversation memory across sessions, Paid LLM fallback (\$5 budget) to eliminate call drops